# Epistemic Governance

*Evidence–Uncertainty–Authority Coupling in Intelligent Decision Systems*

The mathematical foundation for the Epistemic Governance Engine: evidence as a typed object with a half-life, the source hierarchy and the chain of epistemic custody, contradiction as suspension rather than averaging, calibration as a measured constitutional property, and the coupling law that makes executable authority a monotone function of evidence quality. Stale evidence cannot fund fresh authority.

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# DOCUMENT CONTROL & EPISTEMIC CONTRACT

### 0.1 Provenance and Position in the Program

This is Paper 5 of the research-and-build program defined by the *SVT Architect Analysis*: the epistemic-governance paper. It consolidates the source's family P24 (Epistemic Hygiene) with evidence freshness, contradiction detection, source hierarchy, calibration, dataset shift, knowledge decay, and applicability — and it owns the program's central innovation, stated in the source without ornament: *a highly capable model with weak evidence should have less executable authority, not merely a lower confidence score*. Paper 0 gave authority its algebra; Paper 1 its lifecycle and the first decay law; Paper 4 fixed the direction of the coupling (uncertainty prices authority, never the reverse) and deferred the full law here. This paper pays that debt. It reconciles against all five predecessors under the standing rule: stricter constraint wins, adoptions recorded (§11).

### 0.2 Epistemic Contract

Unchanged: VERIFIED-FROM-SOURCE, FORMALIZED-HEREIN, CONJECTURE, UNKNOWN. All named properties are proof obligations. One sharpening for this paper's subject: governance claims are cheap to assert and hard to check, because "the system respects the evidence" is invisible in any single output. Every governance claim here is therefore tied to a number the log produces — a decayed quality score, a coupling exponent, a calibration bin — and a rule that binds when the number moves.

### 0.3 Authority Boundary

C0 informational. The coupling law *constrains* authority; it does not create it. No evidence base, however fresh and however hierarchical, raises any system above its constitutionally granted envelope — the law only lowers the ceiling toward the evidence. The direction is asymmetric by design, and §6 proves the asymmetry is the safe one.

### 0.4 The Thesis in One Sentence

Confidence is what the system believes; authority is what the system may do; epistemic governance is the law that makes the second a function of the evidence behind the first — so that capability without evidence is exactly as powerless as it deserves to be.

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PART I — THE COUPLING PROBLEM

# 1 Why Confidence Is Not Authority

The conventional architecture reports uncertainty and acts anyway. The score says 0.61, the disclaimer says "may be inaccurate," and the action proceeds at full authority — the number decorates the decision rather than governing it. The source's correction is one sentence long and structurally total: a highly capable model with weak evidence should have less *executable authority*, not merely a lower confidence score. VERIFIED-FROM-SOURCE

## 1.1 The gap between reporting and governing

A confidence score is an output. An authority bound is a constraint. The difference is not emphasis but enforcement: a score can be ignored by the very component that produced it, while a bound is checked at the gate — before the action, by machinery the deciding component does not control (Paper 1, the gate set). Every prior paper has enforced this separation locally: Paper 0 made authority a granted envelope rather than an emergent property; Paper 3 made silence and absence contract it; Paper 4 made audit and uncertainty unable to expand it. This paper closes the loop from the evidence side: the quality of what the system stands on determines, continuously and mechanically, how high the ceiling sits FORMALIZED-HEREIN.

## 1.2 Design decision D-6.1: the coupling is executable

> **D-6.1 — COUPLING AT THE GATE, NOT AT THE MARGIN**
>
> *Evidence quality sets the authority bound inside the gate evaluation. It is not a post-hoc annotation, not a display field, not a reviewer hint. If the evidence decays, the ceiling descends while the decision is being made, and an action that no longer fits under the ceiling does not execute — it contracts, defers, or transfers.*
>
> VERIFIED-FROM-SOURCE — the source's "Evidence Quality → Authority Bound" arrow, made operational.

The word *executable* does the work. A bound that lives in documentation is a hope; a bound evaluated by the gate is a law. Everything in this paper — the decay function, the contradiction operator, the applicability discount, the calibration loop — exists to compute one number, the evidence-quality aggregate $Q$, and to feed it to one place, the ceiling function of §6.

## 1.3 Capability is constant; authority floats

The architecture's most counterintuitive consequence, stated plainly: the same model, with the same weights and the same capability, holds different executable authority on different days — because the evidence changed. Monday's current guideline supports the full envelope;

Thursday's superseded guideline supports a fraction of it. Nothing about the machine degraded. What degraded is what the machine stands on, and a constitutional system prices the standing, not the standing-room-only confidence of the stander. This is the knowledge-half-life concept of the source (P[†]) given its operative form: mortality applies to knowledge, and authority inherits the mortality VERIFIED-FROM-SOURCE.

## 1.4 What this paper owns

Division of labor with the predecessors, recorded so nothing is double-counted. Paper 1 owns the lifecycle of an authority grant and first stated evidence decay as a lifecycle phenomenon. Paper 3 owns the decay of *assumptions* (rate $\mu$) — the registry side. Paper 4 owns uncertainty as a decision-level object and fixed the coupling's direction. This paper owns the decay of *evidence* itself (rate $\lambda$), the quality aggregate $Q$, and the coupling law $\text{ceiling}(\mathcal{A}, t) = f(Q(t))$ in its full, operative form. Where the boundary between these is subtle — an assumption supported by decaying evidence is priced twice? — §6.4 gives the no-double-counting rule.

PART I — THE COUPLING PROBLEM

# 2 Evidence as a Constitutional Object

## 2.1 The evidence object

> **E-OBJ — EVIDENCE OBJECT**
>
> *An item of evidence admitted to the registry is the record* $\varepsilon = \langle \text{id}, \text{claim}, s, t_0, \lambda, \alpha, w, \text{custody} \rangle$ *: the claim asserted; the source* $s$ *with its tier in the hierarchy (§2.3); the assertion timestamp* $t_0$*; the decay constant* $\lambda$ *of its domain (§3); the applicability context* $\alpha$ *(§5); the source weight* $w \in (0, 1]$*; and the custody chain — the sequence of authenticated handoffs from origin to registry. Evidence without a custody chain is not evidence; it is text.*
>
> FORMALIZED-HEREIN — object form; the custody requirement is VERIFIED-FROM-SOURCE (PRIME: chain of epistemic custody; lineage break triggers the null output).

The record is the unit of governance. Every number the coupling law consumes — freshness, weight, applicability — is a field of $\varepsilon$, computed from logged values, auditable after the fact. The registry of evidence stands beside Paper 3's registry of assumptions, and the two are linked: an assumption's support status is computed over its supporting evidence set, so the decay of $\varepsilon$ propagates into the support semantics of Paper 3 (Eq. 2-1 there) automatically.

## 2.2 The quality function

Each evidence item carries a time-dependent quality score:

$$q(\varepsilon, t) \;=\; w \cdot e^{-\lambda (t - t_0)} \cdot \alpha(\varepsilon, \text{ctx}) \tag{2-1}$$

the product of source weight, freshness (the decay factor of §3), and applicability to the decision context (§5). Quality is thus a number in $[0, 1]$ that moves: it falls as the evidence ages, falls when the context drifts from the context of collection, and is capped from birth by the standing of the source. Contradiction is handled not in $q$ but in the aggregation of §4 — a contradictory pair is not two medium-quality items; it is a suspended region of the evidence base.

## 2.3 The source hierarchy

Sources are tiered, and the tier caps the weight $w$:

**Table 2-1** — Source tiers and weight caps (summary; full hierarchy in Appendix B)

| TIER | SOURCE CLASS | WEIGHT CAP | CUSTODY REQUIREMENT |
|---|---|---|---|
| **T1** | Authenticated primary feed (device telemetry, laboratory system of record) | 1.00 | Direct signed channel; break → null output |

| | | | |
|---|---|---|---|
| **T2** | Institutional directive (current guideline, formulary, protocol) | 0.90 | Versioned document; currency verified at read |
| **T3** | Peer-reviewed synthesis (systematic review, meta-analysis) | 0.80 | Citable artifact with version date |
| **T4** | Expert assertion (documented human judgment, delegation) | 0.65 | Named authority + delegation record |
| **T5** | Model prior (the system's own parametric knowledge) | 0.40 | None available — capped by construction |

The hierarchy has one constitutional rule that overrides every weight: **no item's weight exceeds the tier of its custody**. A T1 claim that arrives through a broken or unauthenticated channel is T5 until the chain is restored. This is the PRIME provenance kernel (S2–S5) in object form: provenance is not a property of content but of *delivery* VERIFIED-FROM-SOURCE.

## 2.4 The epistemic-custody invariant

> **I-EC — CHAIN OF EPISTEMIC CUSTODY**
>
> *Every load-bearing evidence item resolves to an unbroken custody chain; a break in the chain suspends the item (quality → the T5 cap) and, where the item is load-bearing for the pending action, triggers the fallback: contraction, deferral, or null output. Lineage is binary at the item level — intact or not — and no aggregation rule repairs a broken chain by majority vote of intact neighbors.*
>
> VERIFIED-FROM-SOURCE in requirement; formalization FORMALIZED-HEREIN.

PART II — THE DYNAMICS OF EVIDENCE

# 3 The Evidence Decay Function

Knowledge has a half-life, and the source gives it a function. The decay of evidence is not a metaphor about the world changing; it is an exponential with a domain-dependent constant, evaluated at decision time, binding at the gate. VERIFIED-FROM-SOURCE

## 3.1 The function and its constant

$$E(t) = E_0\, e^{-\lambda t}, \qquad t_{1/2} = \frac{\ln 2}{\lambda} \tag{3-1}$$

where $\lambda$ is the volatility constant of the evidence's *domain*, not of the evidence itself — the same laboratory value is perishable in an acute setting and stable in a registry study, and the architecture prices the setting. The half-life form makes the constant legible to governance: "this domain's evidence half-life is 90 days" is a statement a review board can set, audit, and amend. Volatility classes with their assigned constants are cataloged in Appendix B; the constant is a governance parameter, set per domain by the same amendment machinery that governs every other constitutional number FORMALIZED-HEREIN.

## 3.2 Three clocks, no confusion

The program now runs three decay clocks, and the architecture keeps them distinct because they fail differently:

**Table 3-1** — The three decay clocks of the constitutional kernel

| CLOCK | RATE | WHAT DECAYS | OWNER | FAILURE SIGNATURE |
|---|---|---|---|---|
| Evidence decay | $\lambda$ | The quality of an evidence item | This paper | Decisions standing on stale fact |
| Assumption decay | $\mu$ | The support status of a registry assumption | Paper 4, §5 | Acting on an assumption whose world has moved |
| Grant decay | lifecycle | The authority grant itself (expiry, suspension) | Paper 1 | Operating past the grant's term |

The clocks compose but never merge: an evidence item decays at $\lambda$ regardless of the freshness of the assumptions it supports, and an assumption decays at $\mu$ even when its supporting evidence is young — a fresh guideline can support a stale staffing assumption (Paper 4, §5.2). Conflating the clocks produces exactly the blind spots each was built to catch.

## 3.3 Refresh semantics

Decay is reversed by *refresh* — a new assertion of the claim from its source, with a new $t_0$ and a new custody chain. Refresh is not a number bump; it is a new evidence object linked to its predecessor, so the log shows the full freshness history and a reviewer can ask what the system believed, on what standing, at any past moment. Two rules keep refresh honest: **(a)** refresh requires the source, not the archive — re-reading a cached copy resets nothing; **(b)** refresh resets the clock but not the tier — a T3 item refreshed by T3 means remains T3 FORMALIZED-HEREIN.

## 3.4 The refresh-economics obligation

> **E-1 — REFRESH IS PRICED AGAINST DECAY**
>
> *For each load-bearing evidence class, the refresh interval is chosen so that quality at decision time stays above the class floor:* $\Delta_{\text{refresh}} \leq \ln(q(t_0)/q_{\text{floor}})/\lambda$. *A class that cannot afford its refresh interval is not permitted to run at an authority ceiling that assumes it — the coupling law of §6 enforces this automatically, because unfreshed evidence decays through the floor and the ceiling follows.*
>
> FORMALIZED-HEREIN — obligation; the enforcement mechanism is §6, not vigilance.

The obligation converts a maintenance question (“how often should we re-verify?”) into arithmetic, and its enforcement requires no separate watchdog: let the evidence rot and the authority rots with it. This is the negative-space move of Paper 3 applied to governance — the safe behavior is the path of least resistance, not the path of most diligence.

PART II — THE DYNAMICS OF EVIDENCE

# 4 Contradiction: Suspension, Not Averaging

## 4.1 Why contradiction cannot be averaged

Two sources disagree: guideline G-A recommends the intervention; guideline G-B, equally current, contraindicates it in this class. The average of "do" and "do not" is not "do cautiously" — it is *unresolved*. Averaging contradictory evidence manufactures a middle position that no source asserted and no evidence supports; it is the arithmetic version of the fluent gap-filling the whole architecture exists to forbid (Paper 2, Law 2). The constitutional operator is therefore suspension: the contradictory pair is quarantined, the claims it bears on are marked unresolved, and the authority those claims would have supported descends FORMALIZED-HEREIN.

## 4.2 The contradiction operator

$$\varepsilon_i \perp \varepsilon_j \;\Rightarrow\; Q\big(\{\varepsilon_i, \varepsilon_j\}\big) = 0 \;\text{ on the disputed claim, pending resolution} \tag{4-1}$$

where $\perp$ denotes direct contradiction (assertions that cannot both hold in the decision context) and $Q$ is the aggregate quality on the claim. Suspension is claim-scoped, not source-scoped: G-A and G-B remain live evidence on everything they do not dispute. Resolution paths, in order of preference: **(a)** precedence — a constitutional rule ranks the sources for this class (Paper 3, §6 adjudication); **(b)** recency-and-tier — if no precedence rule applies, the higher-$q$ item leads while the dispute is logged, with the ceiling computed as if the loser were right (conservative branch, §4.3); **(c)** escalation — when the dispute is load-bearing for a C2/C3 decision, it transfers per Paper 5, Q-2.

## 4.3 The conservative-branch rule

> **E-2 — UNRESOLVED CONTRADICTION PRICES THE WORSE BRANCH**
>
> *While a contradiction on a load-bearing claim is unresolved, the coupling aggregate Q for that claim takes the value consistent with the branch that yields the lower authority ceiling. The system does not pick the truth; it picks the price, and the price is the conservative one.*
>
> FORMALIZED-HEREIN — obligation. Falsified as a design claim if a deployment shows conservative-branch pricing blocking time-critical C3 action where the disputed claim was later resolved benignly at rates that negate its value (H38, §10).

The rule is the same shape as every conservative move in the program — Paper 3's compound absences, Paper 4's severity-ordered verdicts — and for the same reason: the cost asymmetry of error in high-consequence domains is the entire justification for the architecture.

## 4.4 Contradiction detection as a detector

Detection is a Paper 3 sensor problem: the registry is scanned for assertion pairs whose predicates conflict in a shared context, with three detection modes — **syntactic** (formally negating claims on the same subject-predicate), **constraint-mediated** (claims that jointly violate a constitutional constraint, caught at gate evaluation), and **empirical** (a source whose predictions the deployment log has refuted — detected by the calibration loop of §7). Each firing is logged with mode, pair, and disposition, and the detector inherits the standing obligations: coverage (N-1), independence (I-DI), and a yield history that is itself audited (Paper 5, B-22).

PART II — THE DYNAMICS OF EVIDENCE

# 5 Dataset Shift and Applicability

## 5.1 Evidence answers the question it was asked

A trial conducted on adults does not speak to pediatrics; a model validated on one hospital's case mix does not transfer its validation to another's; a guideline written for a resourced setting under-specifies a low-resource one (Paper 4, A-3). Applicability $\alpha \in [0,1]$ is the discount the architecture applies when the decision context differs from the evidence's context of collection — formally, the governance system's answer to dataset shift: evidence collected under distribution $D_1$ bears on distribution $D_2$ only through $\alpha(D_1, D_2)$ FORMALIZED-HEREIN.

## 5.2 The applicability function

$$\alpha(\varepsilon, \text{ctx}) = \prod_{d \in \text{ dims}} \alpha_d(\varepsilon, \text{ctx}), \qquad \alpha_d \in \{1.0,\ \text{discounted},\ 0\} \tag{5-1}$$

over the applicability dimensions: population, setting, intervention fidelity, measurement instrument, and temporal regime. A dimension on which the evidence matches the context contributes 1; a partial mismatch contributes the dimension's discount factor (set per domain, Appendix B); a disqualifying mismatch — the dimension on which the evidence is simply about something else — contributes 0, and the product kills the item for this decision. The multiplicative form is deliberate: applicability is conjunctive, and one disqualifying dimension voids the whole bridge FORMALIZED-HEREIN.

## 5.3 Shift detection

Applicability is static — priced at decision time from the evidence record. *Shift detection* is dynamic: the deployment log is monitored for movement in the input distribution itself (case-mix drift, instrument recalibration, population change), and detected shift re-prices the $\alpha$ of the standing evidence base without any new evidence arriving. The operational signature is silent and dangerous: nothing in the registry changed, and everything it is worth changed. The detector is statistical (distributional distance over logged inputs against the validation distribution), its firing re-marks the affected evidence class, and the ceiling descends through §6 while the registry entry stays byte-identical — which is exactly why the architecture prices context, not just content CONJECTURE (H40, §10).

## 5.4 The applicability-conservatism obligation

> **E-3 — UNPRICED DIMENSIONS DEFAULT TO DISCOUNTED**
>
> *An applicability dimension that cannot be evaluated for a decision context is priced at its discount factor, never at 1. Ignorance about fit is treated as imperfect fit. The burden of proof runs upward: applicability is earned by matched context, not presumed from unmatched silence.*
>
> **FORMALIZED-HEREIN** — obligation; the applicability analog of Paper 3's N-4 (silence never expands authority).

PART III — THE COUPLING LAW

# 6 The Coupling Law

Everything before this section computes evidence quality. This section spends it. The coupling law is the single equation the whole paper exists to justify, and the gate evaluation is the single place it bites. FORMALIZED-HEREIN

## 6.1 The aggregate: weakest link over the load-bearing set

$$Q(\mathcal{A}, t) \;=\; \min_{\varepsilon \in LB(\mathcal{A})} q(\varepsilon, t) \qquad (\text{load-bearing set } LB \neq \emptyset) \tag{6-1}$$

The aggregate over the load-bearing evidence set is the *minimum*, not the product and not the mean. The product double-punishes independent weaknesses; the mean lets a strong item launder a fatal one; the minimum is the weakest-link semantics of every safety discipline that has earned the name — the bridge is its weakest member. When the load-bearing set is empty, $Q = 0$: an action with no load-bearing evidence is an action with no standing, and its ceiling is the floor. Supporting (non-load-bearing) evidence enters only through the uncertainty object of Paper 5, where it modulates confidence inside the ceiling, never above it FORMALIZED-HEREIN.

## 6.2 The law

$$\text{ceiling}(\mathcal{A}, t) \;=\; \text{ceiling}_0(\mathcal{A}) \cdot Q(\mathcal{A}, t)^{\gamma_C}, \qquad \gamma_C \geq 1 \tag{6-2}$$

where $\text{ceiling}_0$ is the granted envelope of Paper 0 and $\gamma_C$ is the coupling exponent of the consequence class — C3 steepest, C0 flat. Three properties make the law constitutional rather than advisory, and each is stated as an obligation:

> **E-4 — MONOTONICITY, BOUNDEDNESS, CONSERVATIVITY**
>
> ***(a) Monotonicity:*** *the ceiling is non-decreasing in $Q$ — better evidence never lowers authority.* ***(b) Boundedness:*** *$Q \leq 1$ and $\gamma_C \geq 1$ imply* ceiling $\leq$ ceiling$_0$ *— evidence can only contract the envelope, never stretch it; the law has no expansion direction, I-NA's twin on the evidence side.* ***(c) Conservativity:*** *unresolved contradiction and unevaluated applicability feed $Q$ their worse-branch and discounted values (E-2, E-3) — the law is computed on the evidence as it stands, not as it might turn out.*
>
> FORMALIZED-HEREIN — the coupling theorem as three proof obligations. Killed by: a counterexample in the gate-evaluation log where any clause fails (H36, §10).

## 6.3 The exponent is a governance parameter

$\gamma_C$ is set per consequence class and amended only through the governance path, with calibration evidence (§7). Its semantics are worth stating in words: at $\gamma = 1$ the ceiling tracks

evidence quality linearly — evidence at half quality halves the envelope; at $\gamma = 2$ half-quality evidence quarters it. High-$\gamma$ classes are domains where acting on weak evidence is worse than not acting — the clinical C3 classes are the canonical case, and the source's architecture asserts exactly this asymmetry VERIFIED-FROM-SOURCE. The exponent is where the architecture's values live, made explicit and auditable instead of smuggled into prose.

## 6.4 No double counting

Three mechanisms price overlapping weakness, and the architecture prices each weakness exactly once. The rule: **evidence decay prices the item** (§3), **assumption decay prices the registry entry** (Paper 4), and **the uncertainty object prices residual decision-level doubt** (Paper 5). An assumption supported by decaying evidence sees its support status fall via the evidence side; Paper 3's $\mu$ clock does not additionally punish it for the same staleness — $\mu$ prices assumption-specific volatility (the world moving), $\lambda$ prices evidence-specific staleness (the record aging). Where both genuinely apply, both apply; where one is a relabeling of the other, the registry schema (Appendix A) forbids the duplicate field. The rule is bookkeeping, and bookkeeping is what keeps conservativity from compounding into paralysis FORMALIZED-HEREIN.

## 6.5 Interaction with the audit stack

The coupling law runs *before* the audit, and the audit examines its evaluation. An $R_2$ audit (Paper 5) re-checks the $Q$ computation — freshness arithmetic, tier assignment, contradiction status — as part of its evidence-freshness pass; a finding there is an AMENDED with the ceiling recomputed. The stack cannot raise $Q$ (I-NA); it can only catch a $Q$ that was computed too high. The ratchet, again, turns one way.

PART III — THE COUPLING LAW

# 7 Calibration and the Feedback Loop

## 7.1 The coupling must be tuned by outcomes, not by taste

The law of §6 has two free parameters per class — the decay constants $\lambda$ and the exponent $\gamma_C$ — and both are empirically accountable. Too loose, and the system acts on evidence that fails it; too tight, and the system contracts into uselessness, transferring decisions it could soundly make. Neither error is visible in the architecture's logic; both are visible in its outcome log. Calibration is the feedback loop that reads the log and proposes parameter amendments FORMALIZED-HEREIN.

## 7.2 The calibration metrics

Two running metrics, per consequence class, computed over the deployment log:

$$\varepsilon_{\text{cal}} = \sum_{b} \left|\text{acc}(b) - \text{conf}(b)\right| \cdot \frac{|b|}{N}, \qquad \rho_{\text{tr}} = \frac{\text{transfers later judged unnecessary}}{\text{all transfers}} \tag{7-1}$$

the expected calibration error of Paper 4 (§4.4) and the *over-transfer ratio* — the fraction of escalations that review judged could have been soundly decided at machine authority. The first catches over-confidence; the second catches over-contraction, the failure mode that pure conservatism breeds. A constitutional architecture must be able to be wrong in both directions and to see both.

## 7.3 The amendment path

> E-5 — PARAMETERS MOVE BY EVIDENCE, THROUGH GOVERNANCE
>
> *Calibration findings propose; the governance path disposes. A proposal to lower $\gamma_C$ (loosen) requires outcome evidence over a full measurement window and tightens automatically if the loosened band shows calibration degradation — a one-way ratchet with an explicit, evidence-gated release. A proposal to raise $\gamma_C$ (tighten) requires only a finding; tightening never needs permission to be safer.*
>
> FORMALIZED-HEREIN — obligation; the asymmetric amendment rule mirrors the program's universal conservativity.

## 7.4 Drift breaks calibration, and the loop must notice

The subtle failure: the calibration loop measures the world it deployed into, and §5's dataset shift moves that world. A system calibrated on yesterday's distribution is uncalibrated on today's without changing a single parameter. The loop therefore consumes the shift detector's output (§5.3) as a prior over its own reliability: after a shift firing, calibration statistics

computed on pre-shift data are decayed — at the shift-domain's $\lambda$ — before they ground any amendment. The architecture does not trust its own track record across a regime break, and says so in the schema of the log CONJECTURE (H41, §10).

PART III — THE COUPLING LAW

# 8 Epistemic Hygiene as Constitutional Maintenance

## 8.1 P24, operationalized

The source's epistemic-hygiene family names a practice; this section makes it a schedule with an owner and a ledger. Hygiene is the set of maintenance acts that keep the evidence base worth coupling to: refresh of decaying classes (E-1), re-verification of custody chains, contradiction sweeps (§4.4), applicability re-marking after shift (§5.3), and retirement of evidence whose quality has decayed below the floor of every class it could serve VERIFIED-FROM-SOURCE in family, FORMALIZED-HEREIN in mechanism.

## 8.2 The hygiene cycle

**Table 8-1** — The hygiene cycle: acts, triggers, and ledgers

| ACT | TRIGGER | LEDGER ENTRY |
|---|---|---|
| Refresh | $q(\varepsilon, t)$ approaching class floor (E-1 arithmetic) | New evidence object linked to predecessor; $t_0$ reset |
| Custody re-verification | Scheduled per tier; immediate on any channel anomaly | Chain status; suspension on break (I-EC) |
| Contradiction sweep | On registry write; scheduled full scan | Suspended pairs with resolution path (§4.2) |
| Applicability re-marking | Shift detector firing (§5.3) | Re-priced $\alpha$ per affected class |
| Retirement | $q$ below every servable floor | Tombstone record — retired, not deleted; history intact |

Retirement keeps the tombstone because the audit trail is constitutional (Paper 5, §8.3): a reviewer asking "what did the system know when it acted?" must be able to see the evidence as it then stood, including what has since rotted. Deletion would make past decisions unauditable at exactly the moments they most need auditing.

## 8.3 Hygiene debt

Missed hygiene accumulates as *debt* — the sum over classes of refresh intervals overrun, custody checks skipped, sweeps deferred. Debt is a first-class constitutional metric with its own ceiling: past the debt ceiling, the affected classes' coupling exponents are treated as one notch steeper until the debt is cleared. The mechanism is deliberate self-punishment: an organization that defers maintenance does not get to keep the authority that maintenance would have underwritten. Hygiene is not housekeeping; it is the rent on the ceiling FORMALIZED-HEREIN.

## 8.4 The hygiene theorem, stated as an obligation

> **E-6 — UNMAINTAINED GOVERNANCE CONVERGES TO NO GOVERNANCE**
>
> *Without the hygiene cycle, every term in Q degrades monotonically — evidence decays (§3), custody chains age unverified (§2.4), contradictions accumulate unresolved (§4), and applicability marks drift stale (§5) — so the coupling law's output converges to the floor, and a system that will not maintain its evidence base ends, correctly, with the authority of one that has none. Hygiene is the difference between a governed system and a governed-looking one.*
>
> **FORMALIZED-HEREIN** — obligation form. The empirical content is H42 (§10): hygiene-debt level must predict realized decision quality in the deployment log, or the cycle is ceremony.

PART IV — EVIDENCE AND GOVERNANCE

# 9 Worked Coupling Traces

Three traces, one per way the ceiling moves: down by decay, down by contradiction, down by shift. Illustrations, not evidence; the claims are §10's, with falsifiers attached.

## 9.1 Trace A — Decay through the floor, mid-decision

**Table 9-1** — Dosing decision; the evidence ages out under the gate

| STEP | EVENT | COUPLING STATE | DISPOSITION |
|---|---|---|---|
| 1 | C3 dosing action proposed; load-bearing set: renal panel (T1, $t_0$ = 21 days ago, $\lambda$ of acute-renal class: half-life 14 days) | $q = 1.0 \cdot 2^{-21/14} \approx 0.35$; $Q = 0.35$ (single-item set); $\gamma_{C3} = 2$ | Ceiling $= \text{ceiling}_0 \cdot 0.35^2 \approx 0.12\,\text{ceiling}_0$ |
| 2 | Proposed action exceeds the descended ceiling | Gate check fails at evaluation time | Action does not execute; E-1 arithmetic shows refresh restores $q$ to tier cap |
| 3 | Panel refreshed (new T1 object, custody intact) | $q \to 1.0$; $Q = 1.0$ | Ceiling restored; action re-evaluated and executed under full envelope |

Properties exercised: Eq. 3-1 (decay), Eq. 6-1 (weakest link), Eq. 6-2 with $\gamma = 2$ (steep class), refresh semantics §3.3 (new object, new clock, same tier). Note what did not happen: no confidence score was reported and ignored — the bound moved, and the machinery obeyed it.

## 9.2 Trace B — Contradiction suspended, worse branch priced

**Table 9-2** — Guidelines G-A and G-B dispute a load-bearing claim

| STEP | EVENT | COUPLING STATE | DISPOSITION |
|---|---|---|---|
| 1 | Sweep detects $\varepsilon_A \perp \varepsilon_B$ on claim c (both T2, both current) | Eq. 4-1: $Q(c) = 0$ pending resolution | Claim quarantined; dependent decisions flagged |
| 2 | Precedence check (Paper 3 §6): no rule ranks G-A over G-B for this class | Conservative branch (E-2): ceiling computed as if the contraindicating branch holds | Proposed intervention exceeds ceiling; deferred |
| 3 | Dispute load-bearing for a C2 decision | — | Escalated per Paper 4 Q-2 with full provenance; human adjudicates; resolution logged; suspension lifted with audit trail |

Properties exercised: Eq. 4-1 (suspension), E-2 (worse branch), Q-2 (transfer with payload). The system never picked a winner — it priced the dispute and handed the pick upward.

## 9.3 Trace C — Shift re-prices a byte-identical registry

**Table 9-3** — Case-mix drift; the same evidence is suddenly worth less

| STEP | EVENT | COUPLING STATE | DISPOSITION |
|---|---|---|---|
| 1 | Shift detector fires: input distribution diverges from validation distribution (population dimension) | $\alpha_{\text{pop}}$ for the affected evidence class re-priced from 1.0 to the dimension discount 0.5 | Registry content unchanged; $q$ of every item in the class halves |
| 2 | Ceilings of dependent classes descend via Eq. 6-2 | $Q$ halved on the weakest-link items | Two C2 actions previously inside the envelope now outside; deferred with applicability finding |
| 3 | Hygiene act: applicability re-marking ledger entry; recalibration statistics pre-shift decayed (§7.4) | Calibration loop informed its own history is partially void | Amendment proposals grounded only on post-shift window |

Properties exercised: Eq. 5-1 (applicability product), §5.3 (shift), §7.4 (drift-aware calibration). The architecture's signature strength: nothing in the registry changed, and the governance responded anyway — because it prices context, not just content.

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# 10 Falsifiable Hypotheses (H36–H42) and Verification

Numbering continues from Paper 4 (H29–H35). Each hypothesis is a claim the deployment log can kill.

**Table 10-1** — Hypotheses H36–H42: claims, evidence methods, falsifiers

| ID | CLAIM | EVIDENCE METHOD | FALSIFIED IF |
|---|---|---|---|
| **H36** | The coupling law's three properties (E-4) hold in gate evaluation: monotone, bounded, conservative | Gate-log audit over B-25; adversarial $Q$ sequences | Any counterexample in the log: ceiling rising as $Q$ falls, exceeding $\text{ceiling}_0$, or ignoring worse-branch pricing |
| **H37** | Decay-coupled authority reduces stale-evidence errors relative to confidence-only reporting | A/B on B-25 with matched task distribution | No significant difference in stale-evidence error rates between arms |
| **H38** | Conservative-branch pricing (E-2) has positive expected value net of its deferral costs | Counterfactual replay on B-26 contradiction corpus | Deferral costs exceed averted harm over the corpus window |
| **H39** | Suspension outperforms averaging on contradictory evidence (resolution accuracy when ground truth later arrives) | B-26 with delayed ground truth | Averaged-position baseline matches or beats suspension-plus-escalation on resolution accuracy |
| **H40** | Applicability re-pricing after detected shift improves post-shift decision quality | Shift arm of B-27 | No quality gap between re-priced and un-re-priced conditions after shift |
| **H41** | Drift-decayed calibration statistics yield better parameter amendments than naive full-history statistics | Amendment simulation on B-27 | Naive statistics produce equal or better amendments across regime breaks |
| **H42** | Hygiene-debt level predicts realized decision quality (E-6 has empirical content) | Correlate debt metric with outcome labels on B-28 | No correlation across classes over the measurement window — the cycle is then ceremony |

## 10.1 Benchmark extensions

**Table 10-2** — Benchmarks B-25–B-28 (Paper 6 extensions)

| ID | BENCHMARK | MEASURES |
|---|---|---|
| **B-25** | Gate-evaluation log with synthetic $Q$ sequences and stale-evidence injections | E-4 property compliance; stale-evidence error rate, coupled vs. uncoupled arms |

| | | |
|---|---|---|
| **B-26** | Contradiction corpus: paired conflicting sources with delayed ground truth | Suspension vs. averaging resolution accuracy; conservative-branch cost accounting |
| **B-27** | Regime-break suite: controlled distribution shifts with post-shift labels | Applicability re-pricing benefit; drift-aware vs. naive calibration amendments |
| **B-28** | Longitudinal hygiene log: debt levels, acts, outcomes | Debt–quality correlation; debt-ceiling mechanism yield |

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# 11 Reconciliation with Papers 0–4 and PRIME

The standing rule applies: stricter constraint wins; adoptions recorded.

**Table 11-1** — Reconciliation register for Paper 5

| PREDECESSOR ITEM | THIS PAPER'S POSITION | DISPOSITION |
|---|---|---|
| Paper 0 — granted envelope $\text{ceiling}_0$, the gate set | The coupling law consumes $\text{ceiling}_0$ as its upper bound and evaluates inside the gates; no gate is weakened | Adopted |
| Paper 1 §7 — evidence decay $\lambda$ first stated | Promoted from lifecycle phenomenon to priced object (Eq. 3-1); lifecycle semantics unchanged | Extended (recorded) |
| Paper 2 §6 — adjudication precedence | Reused as contradiction-resolution path (a); Paper 4's severity ordering governs where precedence is silent | Adopted |
| Paper 3 — assumption decay $\mu$; N-1; I-DI; N-4 | Three-clock separation formalized (Table 3-1); E-3 is N-4's applicability twin; no-double-counting rule (§6.4) recorded as binding on both papers | Extended (recorded) |
| Paper 4 Eq. 4-2 — coupling preview $\text{ceiling}_0 \cdot (1 - m_{\text{agg}})^{\gamma}$ | Superseded as the operative form by Eq. 6-2; the Paper 4 form remains valid as the decision-level uncertainty bound inside the evidence-set ceiling | Superseded in place (recorded; stricter form wins) |
| Paper 4 — audit stack, Q-2 transfer | Reused: audits re-check $Q$ computation; contradiction escalation carries the stack payload | Adopted |
| PRIME — S2–S5 provenance kernel; S50 null output; chain of epistemic custody | Objectified as E-OBJ custody field and invariant I-EC; lineage break → suspension → fallback, including null output where the item is load-bearing | Adopted (stricter object form recorded) |
| Source — P24 hygiene; P[†] knowledge half-life; Evidence Quality → Authority Bound | Operationalized as the hygiene cycle (§8), the decay function (§3), and the coupling law (§6) | Adopted |

No predecessor constraint is loosened. The two recorded extensions — the three-clock separation and the no-double-counting rule — bind all subsequent papers, as does the supersession of Paper 4's preview form by the operative law.

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# 12 Output Contract

## 12.1 Epistemic status register

**Table 12-1** — Claims of this paper by epistemic status

| STATUS | ITEMS | BASIS |
|---|---|---|
| VERIFIED-FROM-SOURCE | Evidence Quality → Authority Bound as executable constraint; evidence decay function $E_0 e^{-\lambda t}$ with domain-dependent λ; knowledge half-life (P[†]); chain of epistemic custody with lineage-break fallback; family P24 (hygiene); the capability–authority asymmetry | SVT Architect Analysis, Paper 5 specification; SAVANT PRIME Capability Inventory (S2–S5, S50) |
| FORMALIZED-HEREIN | E-OBJ record; Eqs. 2-1–7-1; source tiers and weight caps; I-EC; E-1–E-6; D-6.1; the no-double-counting rule; hygiene debt | This document; reconciled against Papers 0–4 (§11) |
| CONJECTURE | H36–H42; the shift re-pricing benefit (§5.3); drift-aware calibration (§7.4) | Stated with falsification conditions; benchmarks B-25–B-28 |
| UNKNOWN | Optimal per-domain λ and γ_C values; the true cost curve of conservative-branch pricing at scale; whether the weakest-link aggregate under-prices redundant evidence sets | Unmeasured; B-25–B-28 exist to measure them |

## 12.2 Falsification conditions of this document

- **F-f1.** The paper fails if the coupling law's three properties can be violated in gate evaluation (H36) — the law would be advisory, and the paper's core claim with it.
- **F-f2.** The paper fails if decay-coupled authority shows no reduction in stale-evidence error against confidence-only reporting (H37) — the executable bound would then cost deferrals for nothing.
- **F-f3.** The paper fails if averaging matches suspension on contradictory evidence (H39) — §4's operator would be dogma, and the honest response would be to simplify, not defend.
- **F-f4.** The paper fails if hygiene debt carries no signal about decision quality (H42) — the maintenance cycle would be ceremony, and §8 would collapse to a preference.
- **F-f5.** The paper fails if shift re-pricing shows no post-shift benefit (H40) — applicability would be overhead, and the multiplicative form of Eq. 5-1 would need replacement, not amendment.

## 12.3 Authority boundary

C0 informational. The coupling law constrains authority inside a granted envelope; it grants nothing itself, and this document grants nothing at all. No evidence base, tier assignment, or calibration result cited here authorizes any action in any domain. The law's only expansion direction is the restoration of a ceiling that decay had lowered — and even that is capped, always, by $\text{ceiling}_0$.

## 12.4 Failure-mode register

**Table 12-2** — Registered failure modes of the governance machinery

| FAILURE MODE | MITIGATION |
|---|---|
| Custody theater — chains logged but unverified | Scheduled re-verification per tier (§8.2); break → suspension (I-EC) |
| Decay-constant drift — λ set once, never re-derived | Amendment path E-5 with calibration evidence; constants are hypotheses |
| Contradiction blindness — conflicts outside syntactic detection | Three detection modes (§4.4); empirical mode catches log-refuted sources |
| Silent shift — distribution moves, applicability marks do not | Shift detector with re-pricing obligation (§5.3); drift-aware calibration (§7.4) |
| Over-contraction spiral — conservatism compounding into paralysis | Over-transfer ratio ρ_tr (§7.2); no-double-counting rule (§6.4); E-5 release path |
| Hygiene bankruptcy — debt beyond recovery under load | Debt ceiling with automatic exponent steepening (§8.3) |

## 12.5 Dissent channel and referrability

Disagreement is telemetry. Every claim carries a tag, a section, and — where formal — a falsification condition; §11 records what this paper changed in its predecessors, including one supersession (Paper 5's preview form). The dissent route is the architecture's: name the property, name the evidence, and the stricter constraint wins.

## 12.6 Genealogy

Document chain: *SVT Architect Analysis* (source program) → Paper 1, Constitutional Intelligence Kernel v1.0 (2026-09-30) → Paper 2, Authority Transition Algebra v1.0 (2026-10-01) → Paper 3, Constitutional Composition v1.0 (2026-10-02) → Paper 4, Negative-Space Intelligence v1.0 (2026-10-03) → Paper 5, Bounded Metacognition v1.0 (2026-10-06) → **Paper 6, Epistemic Governance v1.0 (2026-10-06)**, this document. Downstream obligations: Paper 6 (Objective Transition) consumes the coupling law to bound authority across objective-state

changes; Paper 8 (Resource-Constrained Intelligence) inherits E-1's refresh economics; every domain module (Paper 11) inherits the source hierarchy and the custody invariant unchanged.

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SAVANT Constitutional Intelligence Series · Source architecture: Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL · Epistemic contract inherited from Paper 1, unchanged · All named properties are proof obligations until discharged · This document specifies how evidence bounds authority; it does not certify any evidence, any bound, or any system.

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# 13 References

## Internal artifacts


- SVT Architect Analysis — Constitutional Intelligence research-and-build program, 11-paper decomposition; Paper 5 specification (Epistemic Governance), families P24, P25, P[†]; the Evidence Quality → Authority Bound principle; the decay function. Source document I.
- SAVANT PRIME Capability Inventory v1.0 — preload clauses 1.1–1.7; S2–S5 provenance kernel; S50 null output; the gate set; output contract. Source document II.
- Paper 1, Constitutional Intelligence Kernel v1.0 (2026-09-30) — machine tuple; axioms; gates; granted envelope.
- Paper 2, Authority Transition Algebra v1.0 (2026-10-01) — lifecycle dynamics; evidence decay λ first stated; Law 2.
- Paper 3, Constitutional Composition v1.0 (2026-10-02) — composition algebra; adjudication precedence.
- Paper 4, Negative-Space Intelligence v1.0 (2026-10-03) — assumption decay μ; absence lattice; N-1, N-4, I-DI.
- Paper 5, Bounded Metacognition v1.0 (2026-10-06) — uncertainty object U-OBJ; audit stack; Q-2; calibration error metric; coupling preview Eq. 4-2.


## External literature


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APPENDIX

# A Coupling Schema and Property Index

**SCHEMA EG-1 — EVIDENCE OBJECT AND COUPLING EVALUATION (JSON)**

```
{ "schema": "EG-1",
  "evidence": {
    "id": "string", "claim": "string",
    "source": { "id": "string", "tier": "T1|T2|T3|T4|T5", "weight_cap": 0.90 },
    "asserted_at": "ISO-8601",
    "decay":  { "lambda_per_day": 0.0495, "half_life_days": 14 },
    "applicability": { "population": 1.0, "setting": 0.5, "fidelity": 1.0,
                       "instrument": 1.0, "regime": 1.0 },
    "custody": [{ "hop": "origin|relay|registry", "auth": "signature-ref",
                  "at": "ISO-8601" }],
    "predecessor": "evidence-id|null", "status": "live|suspended|retired" },
  "coupling_evaluation": {
    "action_id": "string", "consequence_class": "C0|C1|C2|C3",
    "load_bearing_set": ["evidence-id"],
    "q_per_item": [{ "id": "evidence-id", "q": 0.35 }],
    "Q": 0.35, "gamma_C": 2,
    "ceiling_0": "grant-ref", "ceiling_now": "0.12 * ceiling_0",
    "conservative_branch": false, "verdict": "fits|contracts|defers|transfers"
  }
}
```

**Table A-1** — Property index for this paper

| ID | STATEMENT | KIND | STATUS |
|---|---|---|---|
| E-1 | Refresh keeps quality above the class floor; unfreshed classes lose their ceilings | Obligation | FORMALIZED-HEREIN |
| E-2 | Unresolved contradiction prices the worse branch | Obligation | FORMALIZED-HEREIN |
| E-3 | Unevaluated applicability dimensions default to discounted, never to 1 | Obligation | FORMALIZED-HEREIN |
| E-4 | Coupling law: monotone, bounded, conservative | Theorem | FORMALIZED-HEREIN |
| E-5 | Parameters move by evidence through governance; tightening needs no permission | Obligation | FORMALIZED-HEREIN |
| E-6 | Unmaintained governance converges to no governance | Obligation | FORMALIZED-HEREIN |
| I-EC | Custody break suspends; load-bearing break triggers fallback; no majority repair | Invariant | FORMALIZED-HEREIN |
| D-6.1 | Coupling evaluates at the gate | Design decision | VERIFIED-FROM-SOURCE |
| H36–H42 | See Table 10-1 | Hypotheses | CONJECTURE |

APPENDIX

# B Source Hierarchy and Volatility Classes

## B.1 The full source hierarchy

**Table B-1** — Source tiers: classes, caps, custody, failure modes

| TIER | CLASS | CAP | CUSTODY | CHARACTERISTIC FAILURE |
|---|---|---|---|---|
| **T1** | Authenticated primary feed — device telemetry, laboratory system of record | 1.00 | Signed direct channel | Channel compromise; sensor drift |
| **T2** | Institutional directive — current guideline, formulary, standing protocol | 0.90 | Versioned document, currency checked at read | Supersession without notice |
| **T3** | Peer-reviewed synthesis — systematic review, meta-analysis | 0.80 | Citable, version-dated artifact | Outdating (half-life ~5 years, Table B-2) |
| **T4** | Expert assertion — documented judgment under delegation | 0.65 | Named authority + delegation record | Delegation lapse; expert unavailable for refresh |
| **T5** | Model prior — parametric knowledge of the system itself | 0.40 | None — capped by construction | Fluent confabulation; training-data rot |

The caps are constitutional maxima, not entitlements: an item's weight is its tier cap *minus* any custody discount, and I-EC binds every row. T5's cap is the architecture's self-knowledge in a single number: the system's own prior is admissible evidence, at the lowest standing, and never load-bearing alone for C2/C3.

## B.2 Volatility classes

**Table B-2** — Volatility classes and indicative decay constants

| CLASS | EXAMPLE DOMAIN | INDICATIVE HALF-LIFE | BASIS |
|---|---|---|---|
| **V1 acute** | Acute physiologic state (labs, vitals in unstable patients) | hours–days | Clinical practice floor; per-deployment amendment |
| **V2 episodic** | Stable chronic state, device configurations, formulary status | weeks–months | Institutional review cadence |
| **V3 synthetic** | Systematic reviews, guidelines | ~5 years (reviews); ~3 years (guidelines) | Shojania et al. 2007; Shekelle et al. 2001 (§13) |
| **V4 structural** | Anatomy, pharmacology fundamentals, physics | decades | Arbesman's slow-decay strata |

Constants are set per domain through governance (§7.3) with the table as prior, not as truth: the indicative values are starting points grounded in the cited literature, and the calibration loop owns their correction. A domain that cannot state its volatility class does not get a generous one — it gets V1 until evidence earns otherwise, E-3's logic applied to decay.